

TIRZEPATIDE

Dual GIP + GLP-1 Receptor Agonist

Complete Injection & Dosing Guide -- All Vial Sizes

Purity: 99.8% | Endotoxin-Free | Heavy Metal-Free | Research Use Only

Vial Size Comparison

Tirzepatide Vial Size Comparison -- SaiyanMed



Vial	BAC Water	Concentration	Doses (2.5mg ea)	Examples per Kit (x10)
5mg x 10	1 mL	5.0 mg/mL	2	20 doses (0.25mg ea) 10 doses (0.5mg ea) 2 doses (2.5mg ea)
10mg x 10	2 mL	5.0 mg/mL	4	40 doses (0.25mg ea) 20 doses (0.5mg ea) 4 doses (2.5mg ea)
15mg x 10	2 mL	7.5 mg/mL	6	6 doses (2.5mg ea) 3 doses (5mg ea)
20mg x 10	2 mL	10.0 mg/mL	8	8 doses (2.5mg ea) 4 doses (5mg ea) 2 doses (10mg ea)
30mg x 10	2 mL	15.0 mg/mL	12	12 doses (2.5mg ea) 6 doses (5mg ea) 3 doses (10mg ea)
40mg x 10	2 mL	20.0 mg/mL	16	16 doses (2.5mg ea) 8 doses (5mg ea) 4 doses (10mg ea)
60mg x 10	2 mL	30.0 mg/mL	24	24 doses (2.5mg ea) 12 doses (5mg ea) 6 doses (10mg ea)
80mg x 10	2 mL	40.0 mg/mL	32	32 doses (2.5mg ea) 16 doses (5mg ea) 8 doses (10mg ea)

WHAT IS TIRZEPATIDE?

Tirzepatide is a **first-in-class dual receptor agonist** that simultaneously activates two key metabolic pathways: **GIP** (Glucose-dependent Insulinotropic Polypeptide) and **GLP-1** (Glucagon-Like Peptide-1) receptors. FDA-approved as Mounjaro (diabetes) and Zepbound (weight management), it has demonstrated clinically proven weight reduction of 15-22% in landmark trials — the SURPASS and SURMOUNT series.

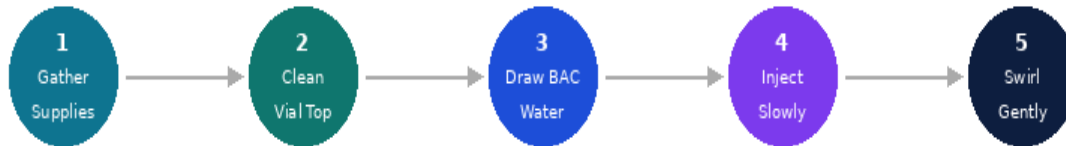
Receptor	Primary Action	Clinical Benefit
GIP	Insulin sensitisation, fat storage regulation, bone metabolism, adiponectin modulation	Amplifies weight loss vs GLP-1 alone; improves lipid profile; fewer GI side effects at equivalent weight-loss doses
GLP-1	Appetite suppression, gastric emptying delay, insulin secretion, glucagon suppression	Reduced caloric intake, improved glycaemic control, HbA1c reduction up to 2.3%

Key Research & Clinical Benefits

Benefit	Evidence & Context
Clinically Proven Weight Loss	SURMOUNT-1 trial: up to 22.5% body weight reduction at 15mg over 72 weeks — the highest weight loss in a randomised trial at time of publication.
Superior Glycaemic Control	SURPASS trials: HbA1c reduction of up to 2.58%, outperforming semaglutide at comparable doses across all SURPASS-2 comparator arms.
Dual Receptor Synergy	GIP + GLP-1 co-agonism produces additive effects on appetite, energy expenditure, and insulin sensitivity beyond either pathway alone.
Cardiovascular Benefits	SURMOUNT-MMO data shows significant reductions in blood pressure, LDL cholesterol, and inflammatory markers alongside weight reduction.
Lean Mass Consideration	Clinical data shows dual agonism preserves lean body mass better than caloric restriction alone; active area of ongoing research.
Liver Fat Reduction	SURPASS data shows significant hepatic fat reduction; potential relevance to NAFLD/NASH research and metabolic liver disease.

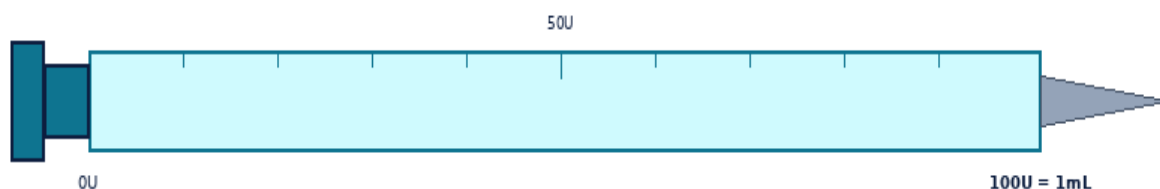
RECONSTITUTION PROTOCOL

Always use **Bacteriostatic Water (BAC Water)** for reconstitution. Allow both the peptide vial and BAC Water to reach room temperature before mixing. Use a fresh insulin syringe for each reconstitution.



Step	Action	Detail
1	Gather supplies	Peptide vial, BAC Water vial, insulin syringes (29-31G), alcohol swabs, sharps container, date labels.
2	Clean vial stoppers	Wipe rubber stoppers of BOTH vials with a fresh alcohol swab. Air-dry for 10-15 seconds before proceeding.
3	Draw BAC Water	Draw the correct volume per vial size (Page 1 table). Expel air bubbles before injection.
4	Inject into peptide vial	Direct the stream against the GLASS WALL — not directly onto the powder. Inject slowly and steadily.
5	Mix gently	Roll vial between palms for 20-30 seconds. NEVER shake — agitation degrades the peptide.
6	Inspect solution	Must be clear and colourless. Discard immediately if cloudy, discoloured, or contains visible particles.
7	Label and refrigerate	Write reconstitution date on vial. Store at 2-8 deg C. Use within 28 days. Never freeze.

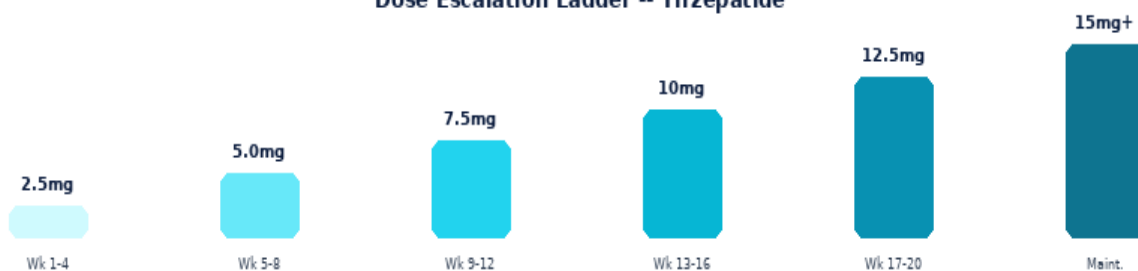
DOSE CALCULATION GUIDE



Insulin Syringe -- 1mL / 100 Units (29-31G needle)

Formula: Volume (mL) = Desired Dose (mg) / Concentration (mg/mL) | **Syringe Units:** Volume (mL) x 100 = Units on insulin syringe

Dose Escalation Ladder -- Tirzepatide



Vial: 5mg x 10 — Add 1 mL BAC Water → Concentration: 5.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.500	1.000	1.500	2.000	2.500	3.000
Syringe Units	50.0U	100.0U	150.0U	200.0U	250.0U	300.0U

Vial: 10mg x 10 — Add 2 mL BAC Water → Concentration: 5.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.500	1.000	1.500	2.000	2.500	3.000
Syringe Units	50.0U	100.0U	150.0U	200.0U	250.0U	300.0U

Vial: 15mg x 10 — Add 2 mL BAC Water → Concentration: 7.5 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.333	0.667	1.000	1.333	1.667	2.000
Syringe Units	33.3U	66.7U	100.0U	133.3U	166.7U	200.0U

Vial: 20mg x 10 — Add 2 mL BAC Water → Concentration: 10.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.250	0.500	0.750	1.000	1.250	1.500
Syringe Units	25.0U	50.0U	75.0U	100.0U	125.0U	150.0U

DOSE CALCULATION GUIDE (CONTINUED)

Vial: 30mg x 10 — Add 2 mL BAC Water → Concentration: 15.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.167	0.333	0.500	0.667	0.833	1.000
Syringe Units	16.7U	33.3U	50.0U	66.7U	83.3U	100.0U

Vial: 40mg x 10 — Add 2 mL BAC Water → Concentration: 20.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.125	0.250	0.375	0.500	0.625	0.750
Syringe Units	12.5U	25.0U	37.5U	50.0U	62.5U	75.0U

Vial: 60mg x 10 — Add 2 mL BAC Water → Concentration: 30.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.083	0.167	0.250	0.333	0.417	0.500
Syringe Units	8.3U	16.7U	25.0U	33.3U	41.7U	50.0U

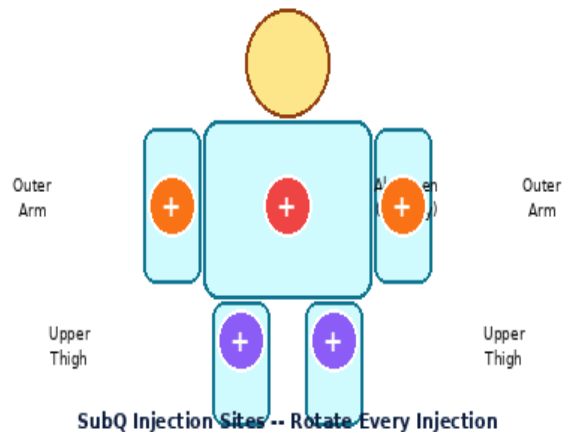
Vial: 80mg x 10 — Add 2 mL BAC Water → Concentration: 40.0 mg/mL

Dose (mg)	2.5mg	5.0mg	7.5mg	10.0mg	12.5mg	15.0mg
Volume (mL)	0.062	0.125	0.188	0.250	0.312	0.375
Syringe Units	6.2U	12.5U	18.8U	25.0U	31.2U	37.5U

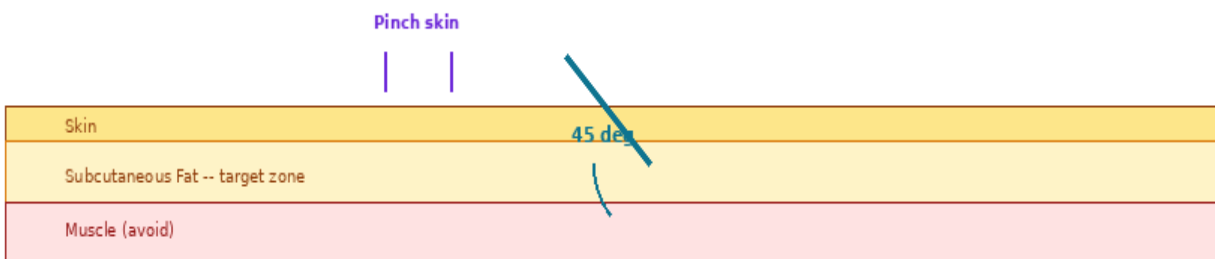
Note: Clinical doses for Tirzepatide are 2.5-15 mg/week. Doses above 15 mg are advanced research doses. Always escalate gradually and under qualified supervision.

INJECTION TECHNIQUE

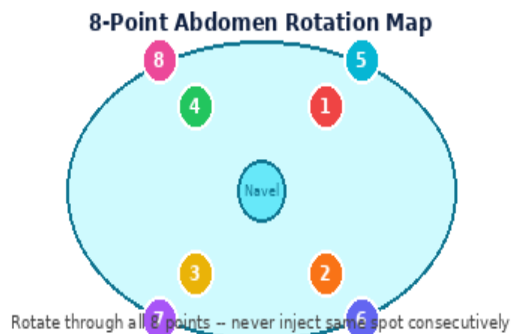
Step	Action	Detail
1	Wash hands	Wash with soap and warm water for 20 seconds. Dry completely.
2	Gather materials	Insulin syringe (29-31G x 5/16"), Tirzepatide vial, alcohol swabs, sharps container.
3	Clean injection site	Wipe chosen site in a circular motion with an alcohol swab. Allow 10-15 seconds to dry completely before injecting.
4	Draw the dose	Draw the correct volume per dose table. Tap and expel all air bubbles before proceeding.
5	Pinch and insert	Pinch 2-3 cm of skin. Insert needle at 45 degrees. Release pinch. Inject slowly over 5-10 seconds.
6	Withdraw and press	Remove at same 45-degree angle. Press site gently 5-10 sec with dry swab. Do NOT rub — rubbing displaces the dose.
7	Safe disposal	Place needle immediately into sharps container. Never recap, reuse, or leave needles exposed.



45-Degree SubQ Injection Angle



SITE ROTATION & DOSING PROTOCOL



Weekly Dosing Schedule -- Same Day Each Week



Choose any consistent day (Mon shown). 1 injection per week only.

Phase	Weeks	Dose	Frequency	Notes
Starter	1-4	2.5 mg	1x / week	Assess GI tolerance. Inject with or just after a meal.
Step 2	5-8	5.0 mg	1x / week	Increase only if 2.5 mg is well tolerated for 4 weeks.
Step 3	9-12	7.5 mg	1x / week	Monitor weight, appetite, blood glucose if applicable.
Step 4	13-16	10 mg	1x / week	Many subjects reach adequate response at this dose.
Step 5	17-20	12.5 mg	1x / week	Escalate if further response desired and tolerating well.
Maintenance	20+	15 mg	1x / week	Max approved clinical dose. Higher doses for research only.
Rest Period	Post-cycle	0 mg	4-8 wk off	Allow receptor sensitivity reset before repeating cycle.

Administration tip: Always inject Tirzepatide with or shortly after a meal — this significantly reduces nausea, especially during the escalation phase.

Consistency: Same day each week (e.g., every Monday). A 1-2 day variance is acceptable; never double-dose to compensate for a missed injection.

SIDE EFFECTS, STORAGE & GOLDEN RULES

Side Effect	Likelihood	Management
Nausea	Very Common	Inject with food. Small meals. Avoid fatty/spicy food on injection day.
Reduced Appetite	Very Common	Expected therapeutic effect. Maintain adequate protein intake.
Diarrhoea	Common	Usually transient. Stay hydrated. Reduce dose if persistent.
Vomiting	Common	Typically dose-related. Hold escalation until resolved.
Constipation	Occasional	Increase fluid and fibre intake. Usually self-resolving.
Injection site reaction	Occasional	Rotate sites systematically. Use fresh 29-31G needle.
Hypoglycaemia (if on insulin)	Occasional	Monitor blood glucose. Adjust concomitant insulin dose as needed.
Tachycardia / Palpitations	Rare	Stop dosing. Seek medical advice before resuming.

STOP & SEEK MEDICAL ADVICE: Severe persistent abdominal pain (rule out pancreatitis)

STOP & SEEK MEDICAL ADVICE: Neck lump, hoarseness, or difficulty swallowing (thyroid concern)

STOP & SEEK MEDICAL ADVICE: Severe allergic reaction: hives, facial swelling, breathing difficulty

STOP & SEEK MEDICAL ADVICE: Resting heart rate increase > 20 bpm above personal baseline

STOP & SEEK MEDICAL ADVICE: Jaundice or dark urine (hepatic concern)

STOP & SEEK MEDICAL ADVICE: Severe or persistent hypoglycaemia (especially if on insulin)

Absolute Contraindications: Personal or family history of medullary thyroid carcinoma or MEN2 syndrome, active or history of pancreatitis, pregnancy or lactation. Not for use in individuals under 18.

Storage Guidelines

State	Temperature	Duration	Notes
Unreconstituted powder	2-8 deg C	Per expiry	Original vial. Dry, away from light.
Reconstituted solution	2-8 deg C	28 days	Label with date. Inspect before each use.
BAC Water (opened)	2-8 deg C	28 days	Label with opening date.
Short-term transport	Below 25 deg C	Max 72 hr	Insulated bag + ice packs. No direct ice contact.

Golden Rules — Tirzepatide

1. Always start at 2.5 mg and escalate in 2.5 mg steps every 4 weeks — no skipping.
2. Inject with or after food to minimise nausea, especially during ramp-up.
3. One injection per week only — more frequent dosing does not improve outcomes.
4. Never inject the same site twice in a row — rotate through all 8 marked points.
5. Use only BAC Water for reconstitution. Never use saline or plain sterile water.
6. Swirl gently to mix — NEVER shake. Agitation denatures the peptide.
7. Label every vial with the reconstitution date. Discard after 28 days.

8. Store at 2-8 deg C at all times. Never freeze reconstituted peptide.

9. Contraindicated in thyroid cancer history, MEN2, pancreatitis, or pregnancy.

10. If GI side effects are severe at a given dose, hold that dose for an extra 4 weeks before escalating.

DISCLAIMER: This document is produced by SaiyanMed for informational and research reference purposes only. Tirzepatide is a research compound in this context. This guide does not constitute medical advice. Always consult a qualified medical professional before beginning any peptide research or weight management protocol. SaiyanMed assumes no liability for misuse of this product or information herein.

SaiyanMed | saiyanmed.com | Research Use Only | For Educational Purposes